

Binary Semantic Segmentation of Surgical Blood

Filippo Giusti, Francesco Marzo

University of Modena and Reggio Emilia

Computer Vision and Cognitive Systems (CVCS) – A.Y. 2025/2026

{300037, 302595}@studenti.unimore.it

August 4, 2026

Abstract

Blood loss is a frequent and time-critical complication during surgery, and being able to automatically localise blood in the surgical field could help warn the surgeon and support part of the intervention. We address this problem as binary, pixel-wise semantic segmentation on two datasets, HemoSet and Rabbani, aiming at a clean and reproducible comparison rather than at a new large architecture. We compare four convolutional baselines—U-Net with ResNet18/34, U-Net++, and DeepLabV3+—in multiclass and binary setups, and find that the binary formulation is consistently better, with U-Net++ leading. Since results are highly sensitive to the data split, we adopt a five-configuration stratified evaluation and report mean and standard deviation. Zero-shot cross-dataset transfer reveals a large, asymmetric performance drop, indicating a substantial domain shift. Finally, we introduce BorDINO, a boundary-aware extension of SegDINO, which markedly improves overlap and boundary accuracy on HemoSet, increasing Dice from 0.7709 to 0.8328 and reducing HD95 from 50.78 to 28.75 pixels, but does not transfer this benefit to Rabbani. Overall, we find that data split and domain shift can outweigh architecture choice, and that boundary-aware supervision is most effective when blood contours are clearly defined.

Keywords: semantic segmentation, surgical blood, HemoSet, Rabbani, robust evaluation, domain shift, SegDINO, boundary-aware learning.

1 Introduction

Bleeding is a common and time-critical event during surgery. Automatically identifying the pixels covered by blood could support intraoperative warning systems and, in the longer term, assist automated hemostasis-management tools. We formulate this task as binary semantic segmentation: for each image pixel, the model predicts either blood or background. We do not distinguish separate blood instances because both datasets considered in this study provide binary masks.

The main objective is to obtain a clean and reproducible analysis of surgical blood segmentation under different modelling and evaluation choices. We first compare established convolutional architectures in multiclass and binary training settings. We then study how strongly the results depend on the train–validation–test partition, repeat the evaluation on a second surgical dataset, and analyse direct transfer across datasets without target-domain adaptation. Finally, we investigate whether a modern frozen visual backbone benefits from explicit contour supervision.

The main contributions of this work are:

- a split-generation procedure that combines blood-content stratification with the grouping constraints available in each dataset;
- a comparison of four convolutional segmentation baselines in multiclass and binary settings, followed by a robust multi-configuration evaluation on HemoSet and Rabbani;
- two complementary zero-shot protocols that expose a large and asymmetric domain shift between the datasets;
- BorDINO, a lightweight boundary-aware extension of SegDINO that improves region overlap, contour accuracy, and split stability on HemoSet without requiring additional manual annotations.

2 Motivation and Background

HemoSet was introduced to support the development of blood-detection systems for robotic hemostasis management and contains annotated frames from live-animal robotic surgery [1]. Rabbani et al. study laparoscopic bleeding management in human surgery and propose a video-based system that combines

temporal modelling and domain adaptation [2]. These datasets address the same nominal class, but they differ in anatomy, acquisition setup, illumination, instruments, bleeding patterns, and annotation style. Such differences make cross-dataset generalisation particularly challenging.

Standard encoder–decoder networks remain strong baselines for medical and surgical segmentation. U-Net recovers spatial information through direct skip connections [3]; U-Net++ uses nested skip pathways to reduce the semantic gap between encoder and decoder features [4]; and DeepLabV3+ combines atrous spatial pyramid pooling with a lightweight decoder to capture multi-scale context [5].

More recently, SegDINO has shown that a frozen DINOv3 backbone can be adapted to segmentation using a lightweight decoder [6]. This design is attractive when the goal is to reuse general-purpose visual representations without fully fine-tuning a large encoder. However, surgical blood often forms thin flows, small regions, irregular pools, and weak transitions against reddish tissue. Region-overlap supervision alone may therefore provide limited information about contour geometry.

Boundary-aware segmentation has been explored in medical imaging by explicitly learning object contours or using them to guide segmentation features [7, 8]. These works motivate our main methodological extension: BorDINO adds an auxiliary boundary-prediction task to SegDINO and tests whether explicit contour supervision helps general-purpose DINO features represent the morphology of blood regions more accurately.

3 Methodology

3.1 Overview

The experimental pipeline has four stages. First, the images are assigned to train, validation, and test sets using dataset-specific splitting rules and blood-content stratification. Second, the selected model is trained using either the multiclass or binary formulation. Third, the validation set is used for checkpoint selection. Finally, the selected checkpoint is evaluated on the held-out test set using overlap and boundary metrics. All model comparisons use the same preprocessing and split configuration within the corresponding experiment.

3.2 Datasets and Splits

HemoSet. HemoSet contains porcine robotic-surgery videos organised by pig identifier, with annotated frames and binary blood masks [1]. Frames close in time within the same video are highly correlated. A random frame-level split could therefore place nearly identical images in training and test. To prevent this leakage, all frames belonging to the same pig/video are kept in a single subset.

Rabbani. The Rabbani dataset contains human laparoscopic frames and binary blood masks [2]. In the local CSV, no reliable patient- or clip-level identifier is available. The splits are therefore created at image level. This preserves the blood-content distribution but cannot guarantee temporal or patient-level independence.

Blood-pixel ratio. To create more balanced and comparable splits, every annotated image is described by the fraction of pixels labelled as blood. For image i , with binary mask Y_i , height H_i , and width W_i , the blood-pixel ratio is

$$r_i = \frac{|\{p : Y_i(p) > 0\}|}{H_i W_i}. \quad (1)$$

A value of $r_i = 0$ represents an empty mask, while larger values indicate that blood occupies a larger fraction of the frame. The continuous ratios are divided into low-, medium-, and high-blood groups using quantile-based bins, so that each group contains approximately the same number of images. Empty masks, when present, are included in the lowest blood-content group.

The goal is to maintain similar proportions of low-, medium-, and high-blood images in train, validation, and test. This reduces the risk that one subset is dominated by large blood regions while another contains mostly small regions.

HemoSet split generation. Ten usable pig/video groups are available in the local copy. An outer **StratifiedGroupKFold** with five folds [9] assigns two pigs to test and leaves eight pigs for train and validation. On these eight groups, an inner **StratifiedGroupKFold** with four folds assigns two pigs to validation and six to training.

The process is repeated with different seeds. Candidate configurations are checked for group disjointness and ranked according to the similarity between their blood-content distributions and the complete dataset. The final five configurations also favour a balanced use of each pig across train, validation, and test roles.

Rabbani split generation. Since reliable groups are unavailable, two consecutive **StratifiedShuffle Split** operations [9] are applied at image level. The first separates 101 test images from 650 train–validation images; the second divides the remaining images into 525 training and 125 validation samples.

Different seeds produce five repeated stratified holdouts. In every configuration, the three subsets are disjoint, although an image may take different roles across configurations.

Data augmentation. Augmentation is applied only to training samples. The pipeline introduces moderate spatial and photometric variation through random cropping, resizing, and brightness changes. Geometric transformations are applied identically to the RGB image and its mask, whereas photometric transformations affect only the image. Validation and test transformations are deterministic.

3.3 Models

We evaluate four convolutional baselines: U-Net with ResNet18 and ResNet34 encoders, U-Net++ with ResNet18, and DeepLabV3+ with ResNet18. All convolutional encoders are pretrained on ImageNet. These architectures represent three common segmentation strategies: direct skip connections, nested skip pathways, and multi-scale atrous context.

We additionally evaluate SegDINO [6]. The model extracts multi-level features from a pretrained DINOv3 Vision Transformer while keeping the backbone frozen. Only the lightweight segmentation decoder and prediction head are trained on the surgical datasets.

3.4 Training Paradigms and BorDINO

Multiclass formulation. The network produces two logits per pixel, one for background and one for blood. A softmax converts them into class probabilities, and training uses cross-entropy loss:

$$\mathcal{L}_{\text{multi}} = \mathcal{L}_{\text{CE}}. \quad (2)$$

Binary formulation. The network produces one blood logit per pixel. A sigmoid is used at evaluation time and the final mask is obtained with threshold 0.5. The training objective combines binary cross-entropy with Dice loss:

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}}. \quad (3)$$

The BCE term provides pixel-wise supervision, while the Dice term reduces the effect of the strong imbalance between blood and background pixels.

BorDINO. We extend SegDINO by attaching a small auxiliary boundary head to a decoder feature map. The segmentation head predicts the full blood region, while the new head predicts its contour. Boundary targets are generated automatically from each binary mask Y :

$$Y_b = \text{dilate}(Y) - \text{erode}(Y). \quad (4)$$

No additional manual annotation is required. The boundary loss combines BCE and Dice terms, and the complete objective is

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda_b \mathcal{L}_{\text{boundary}}, \quad \lambda_b = 0.2. \quad (5)$$

The DINOv3 backbone remains frozen. During inference, the segmentation mask is the primary output; the auxiliary boundary head is used mainly to provide an additional training signal to the shared decoder features.

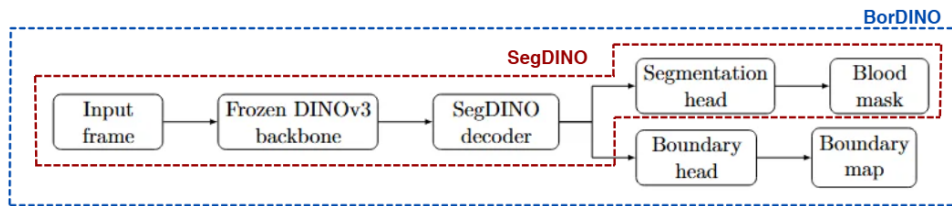


Figure 1: Overview of BorDINO. The dashed blue box identifies the original SegDINO pipeline. BorDINO adds the auxiliary boundary branch while keeping the DINOv3 backbone frozen.

3.5 Metrics

We report complementary region-overlap and boundary metrics. This follows recent recommendations that metric selection and aggregation should reflect the target structure, class imbalance, and hierarchical organisation of the data [10].

For the blood class, intersection over union and Dice are

$$\text{IoU} = \frac{TP}{TP + FP + FN}, \quad \text{Dice} = \frac{2TP}{2TP + FP + FN}. \quad (6)$$

In binary segmentation, Dice is equivalent to the F1-score. Precision and recall are used in the zero-shot analysis to distinguish over-segmentation from missed blood regions:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}. \quad (7)$$

We use two aggregation schemes and define them explicitly because their terminology is not universal.

In the *global*, or dataset-level, scheme, TP , FP , and FN are summed across all test pixels before computing the metric. Images with larger blood regions therefore contribute more.

In the *per-image* scheme, the metric is computed independently for every frame and then averaged, giving each frame equal weight.

For SegDINO and BorDINO, we also report the 95th-percentile Hausdorff distance (HD95), computed from bidirectional distances between predicted and ground-truth contours. Lower HD95 indicates better boundary localisation and is less sensitive to isolated extreme errors than the maximum Hausdorff distance.

4 Experiments

4.1 Multiclass versus Binary Training

We first compare the two training formulations on the same HemoSet split. Table 1 combines the original multiclass and binary results. All four binary variants improve over their corresponding multiclass model, and the binary runs occupy the first four positions in the dataset-level Dice ranking shown in Fig. 2.

Table 1: HemoSet results on the original single split(pig3,pig6,pig11). The table compares global and per-image IoU and Dice under multiclass and binary training. Per-image values are reported as mean $\pm$ standard deviation across test frames.

Model	Encoder	Multiclass				Binary			
		Global		Per-image		Global		Per-image	
		IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice
U-Net	ResNet18	0.6716	0.8036	0.6383 $\pm$ 0.2561	0.7406 $\pm$ 0.2514	0.7153	0.8340	0.6649 $\pm$ 0.2104	0.7750 $\pm$ 0.1920
U-Net	ResNet34	0.6598	0.7950	0.6394 $\pm$ 0.2104	0.7549 $\pm$ 0.1860	0.7059	0.8276	0.6587 $\pm$ 0.2074	0.7710 $\pm$ 0.1906
U-Net++	ResNet18	0.6857	0.8136	0.6401 $\pm$ 0.2106	0.7569 $\pm$ 0.1870	0.7197	0.8370	0.6663 $\pm$ 0.2109	0.7756 $\pm$ 0.1952
DeepLabV3+	ResNet18	0.6817	0.8107	0.6328 $\pm$ 0.2183	0.7486 $\pm$ 0.2020	0.6992	0.8230	0.6500 $\pm$ 0.2111	0.7642 $\pm$ 0.1870

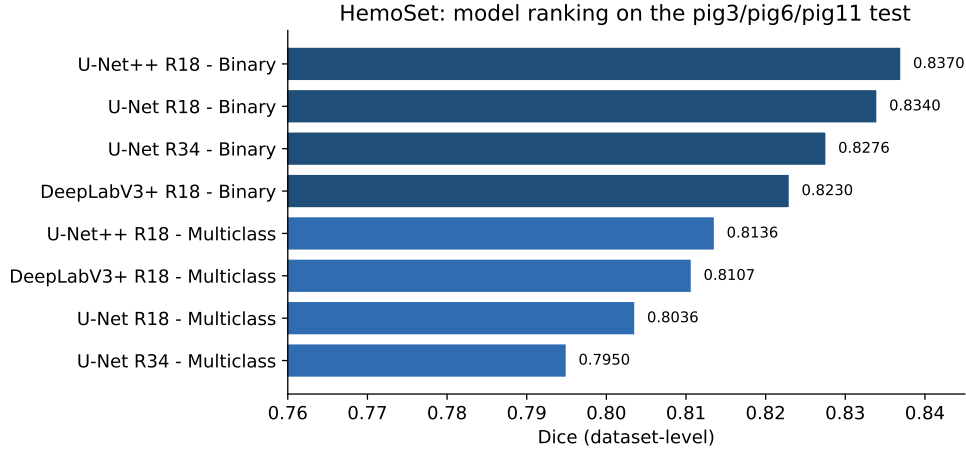


Figure 2: HemoSet model ranking on the original split using dataset-level Dice. The four binary runs rank above the four multiclass runs.

U-Net++ obtains the highest global Dice in both settings. However, its binary score differs from U-Net ResNet18 by only 0.003, so this single split does not support a strong architectural ranking. This observation motivates evaluation across multiple configurations.

4.2 Robust In-Domain Evaluation

Each binary CNN is trained on five configurations. SegDINO and BorDINO are evaluated on three configurations. Table 2 collects the main results in a single table.

The comparison between the CNN and DINO-based families should be interpreted descriptively rather than as a paired statistical test, because they were evaluated over different numbers of configurations. Global metrics were not recorded for the DINO-based experiments, while HD95 was not recorded for the CNN baselines.

Table 2: Main in-domain results. CNN values are mean $\pm$ standard deviation across five configurations; SegDINO and BorDINO values are per-image results across three configurations. Higher IoU and Dice are better; lower HD95 is better.

HemoSet					
Model	Global IoU	Global Dice	Per-image IoU	Per-image Dice	HD95
U-Net R18 ⁽⁵⁾	0.6696 $\pm$ 0.1017	0.7987 $\pm$ 0.0708	0.6191 $\pm$ 0.0937	0.7464 $\pm$ 0.0674	–
U-Net R34 ⁽⁵⁾	0.6607 $\pm$ 0.0957	0.7925 $\pm$ 0.0701	0.6056 $\pm$ 0.0881	0.7363 $\pm$ 0.0669	–
U-Net++ R18 ⁽⁵⁾	0.6790 $\pm$ 0.0850	0.8064 $\pm$ 0.0586	0.6221 $\pm$ 0.0821	0.7479 $\pm$ 0.0602	–
DeepLabV3+ R18 ⁽⁵⁾	0.6733 $\pm$ 0.0988	0.8015 $\pm$ 0.0684	0.6312 $\pm$ 0.0761	0.7561 $\pm$ 0.0531	–
SegDINO ⁽³⁾	–	–	0.6517 $\pm$ 0.0927	0.7709 $\pm$ 0.0779	50.7790 $\pm$ 31.7245
BorDINO ⁽³⁾	–	–	0.7326 $\pm$ 0.0531	0.8328 $\pm$ 0.0441	28.7532 $\pm$ 6.6815

Rabbani					
Model	Global IoU	Global Dice	Per-image IoU	Per-image Dice	HD95
U-Net R18 ⁽⁵⁾	0.6015 $\pm$ 0.0337	0.7507 $\pm$ 0.0262	0.5382 $\pm$ 0.0349	0.6521 $\pm$ 0.0377	–
U-Net R34 ⁽⁵⁾	0.6017 $\pm$ 0.0275	0.7511 $\pm$ 0.0214	0.5344 $\pm$ 0.0235	0.6469 $\pm$ 0.0248	–
U-Net++ R18 ⁽⁵⁾	0.6134 $\pm$ 0.0172	0.7602 $\pm$ 0.0132	0.5547 $\pm$ 0.0271	0.6696 $\pm$ 0.0262	–
DeepLabV3+ R18 ⁽⁵⁾	0.6006 $\pm$ 0.0211	0.7503 $\pm$ 0.0165	0.5650 $\pm$ 0.0215	0.6827 $\pm$ 0.0200	–
SegDINO ⁽³⁾	–	–	0.6139 $\pm$ 0.0318	0.7406 $\pm$ 0.0243	64.6881 $\pm$ 29.3988
BorDINO ⁽³⁾	–	–	0.6112 $\pm$ 0.0255	0.7322 $\pm$ 0.0256	67.2470 $\pm$ 3.6092

Table 3: Per-image SegDINO and BorDINO results for the three evaluated configurations. HD95 is measured in pixels.

Dataset	Model	Configuration	IoU	Dice	HD95
HemoSet	SegDINO	C1	0.5384	0.6760	95.3212
	SegDINO	C2	0.7654	0.8667	33.1621
	SegDINO	C3	0.6512	0.7701	23.8537
	BorDINO	C1	0.6760	0.7836	31.0068
	BorDINO	C2	0.8037	0.8906	35.5734
	BorDINO	C3	0.7181	0.8241	19.6795
Rabbani	SegDINO	C1	0.5734	0.7105	91.8762
	SegDINO	C2	0.6512	0.7701	23.8537
	SegDINO	C3	0.6172	0.7413	78.3345
	BorDINO	C1	0.5754	0.6966	68.7450
	BorDINO	C2	0.6328	0.7561	62.2723
	BorDINO	C3	0.6253	0.7438	70.7236

Among the CNNs, U-Net++ has the highest mean global overlap on both datasets, whereas DeepLabV3+ has the highest per-image IoU and Dice. The differences between CNN architectures are generally smaller than the variation across splits, so no CNN is uniformly superior.

On HemoSet, BorDINO clearly improves over SegDINO. Mean IoU increases from 0.6517 to 0.7326, Dice from 0.7709 to 0.8328, and HD95 decreases from 50.7790 to 28.7532 pixels. The lower standard deviations also indicate reduced split sensitivity.

On Rabbani, SegDINO retains slightly better mean scores. BorDINO does not improve the average HD95, but its much smaller HD95 standard deviation indicates more consistent contour localisation across configurations.

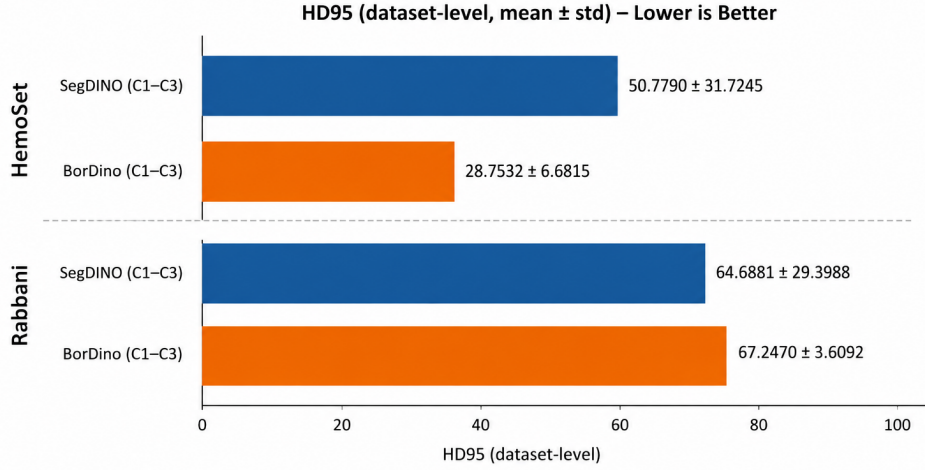


Figure 3: Mean HD95 of SegDINO and BorDINO across three configurations. Lower values indicate better boundary localisation.

The different behaviour may be related to boundary quality. HemoSet often contains more localised pools and visible flows with clearer transitions, whereas blood in Rabbani can be more diffuse, mixed with fluids, or weakly separated from surrounding tissue. Under these conditions, explicit contour supervision may provide a weaker or noisier signal.

4.3 Additional Experiments: Zero-Shot Transfer

We evaluate cross-dataset transfer with U-Net++ because it has the highest mean global Dice among the CNN baselines on both datasets.

Two protocols are considered. In the *paired-configuration* protocol, each checkpoint trained on one source configuration is directly evaluated on the corresponding target test configuration without additional training.

In the *complete-dataset* protocol, a new U-Net++ is trained from scratch using only a train-validation partition of the complete source dataset and is then evaluated on every target image. In both cases, the model never observes target-domain data before final evaluation.

Table 4: Zero-shot cross-dataset transfer. Paired-configuration values are mean $\pm$ standard deviation across five configurations. Complete-dataset per-image values are mean $\pm$ standard deviation across target images.

Global metrics					
Protocol	Direction	IoU	Dice	Precision	Recall
Paired configurations	HemoSet $\rightarrow$ Rabbani	0.2153 $\pm$ 0.0606	0.3512 $\pm$ 0.0787	–	–
Paired configurations	Rabbani $\rightarrow$ HemoSet	0.3194 $\pm$ 0.1110	0.4745 $\pm$ 0.1431	–	–
Complete dataset	HemoSet $\rightarrow$ Rabbani	0.1720	0.2935	0.2958	0.2914
Complete dataset	Rabbani $\rightarrow$ HemoSet	0.3887	0.5598	0.3961	0.9541
Per-image metrics					
Protocol	Direction	IoU	Dice	Precision	Recall
Paired configurations	HemoSet $\rightarrow$ Rabbani	0.1137 $\pm$ 0.0374	0.1640 $\pm$ 0.0508	–	–
Paired configurations	Rabbani $\rightarrow$ HemoSet	0.2837 $\pm$ 0.1111	0.4141 $\pm$ 0.1521	–	–
Complete dataset	HemoSet $\rightarrow$ Rabbani	0.1133 $\pm$ 0.1893	0.1618 $\pm$ 0.2479	0.4316 $\pm$ 0.4410	0.1633 $\pm$ 0.2455
Complete dataset	Rabbani $\rightarrow$ HemoSet	0.3576 $\pm$ 0.1506	0.5081 $\pm$ 0.1708	0.3735 $\pm$ 0.1583	0.9038 $\pm$ 0.1634

Both transfer directions are substantially worse than in-domain evaluation, confirming a strong domain shift. The failure is also asymmetric.

HemoSet $\rightarrow$ Rabbani produces low recall and misses many target blood regions. Rabbani $\rightarrow$ HemoSet reaches very high recall but low precision, indicating substantial over-segmentation of reddish or ambiguous regions.

A model therefore learns not only a general notion of blood, but also dataset-specific relations among colour, texture, context, and annotation style.

4.4 Qualitative Analysis

Quantitative scores should be complemented by direct inspection of the predicted masks. A useful qualitative figure should compare all model families on exactly the same frames, with particular attention to small blood regions, thin flows, irregular boundaries, reflections, surgical instruments, and reddish tissue. Figure 4 compares the main model families on the same surgical frame.

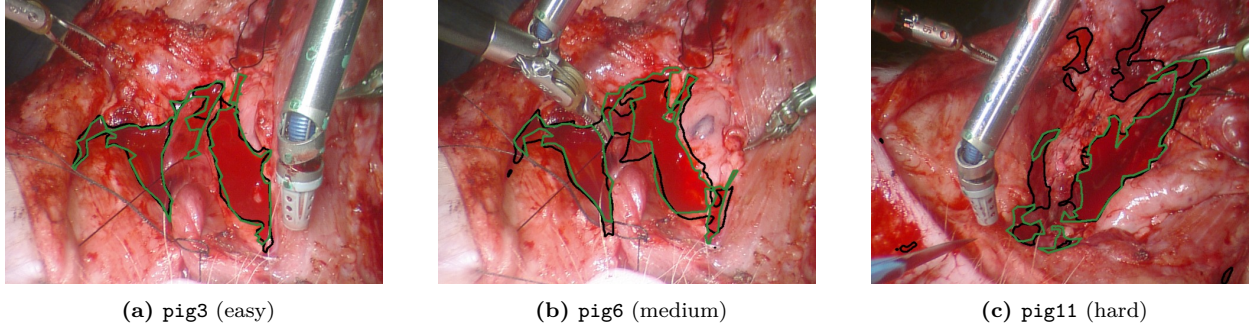


Figure 4: Qualitative segmentation on the HemoSet test videos. **Green** = ground-truth contour, **black** = predicted contour.

5 Conclusion

In this work, we investigated binary semantic segmentation of surgical blood on HemoSet and Rabbani. We found that the binary formulation using BCE and Dice loss consistently outperformed the two-channel multiclass formulation on the original HemoSet split. Across repeated configurations, architectural differences among the CNN baselines were generally smaller than the variation caused by the data split.

The zero-shot experiments revealed a large and asymmetric domain shift: HemoSet-trained models tended to miss diffuse blood in Rabbani, while Rabbani-trained models tended to over-segment ambiguous regions in HemoSet.

Our main methodological contribution, BorDINO, adds auxiliary contour supervision to a frozen-backbone SegDINO model. On HemoSet, this modification produced a clear improvement in overlap, boundary localisation, and stability across splits. The same average benefit was not observed on Rabbani, suggesting that boundary-aware supervision depends on how clearly blood contours are represented and annotated.

The study has several limitations. First, the local dataset copies do not completely match the data described in the original publications. The local HemoSet copy is missing the pig8 folder, reducing the number of available video groups. The Rabbani CSV does not expose a reliable patient- or clip-level identifier. Consequently, the Rabbani configurations can preserve the distribution of blood content but cannot prevent temporally related frames, or frames from the same patient, from appearing in different subsets. Its image-level evaluation is therefore methodologically weaker than the grouped evaluation used for HemoSet.

Second, the experiments show substantial sensitivity to the data partition. The five CNN configurations and the three SegDINO/BorDINO configurations produce variations that are often larger than the differences between architectures. Results obtained from a single split could therefore lead to an optimistic or misleading model ranking. Reporting the individual configurations together with their mean and standard deviation reduces this problem, but a larger number of independent videos and datasets would be required for stronger statistical conclusions.

Third, the original HemoSet and Rabbani publications do not provide enough information to reproduce their exact preprocessing, augmentation, splitting, and checkpoint-selection procedures. Details such as crop size and probability, brightness range, rotation settings, interpolation methods, and transformation order are missing or only described at a high level. We therefore use our own moderate and consistent augmentation pipeline. Although this makes the internal comparison fair, it limits direct comparison with the values reported in the reference papers and may partly explain the observed differences.

Fourth, the cross-dataset analysis is affected by differences that go beyond image appearance. HemoSet contains porcine robotic-surgery scenes in which blood often appears as localised pools, irregular accumulations, or visible flows. Rabbani contains human laparoscopic scenes with different anatomy, instruments, illumination, colour distribution, and bleeding patterns. The two datasets may also follow different annotation conventions, especially for diffuse blood, blood mixed with other fluids, and weak or ambiguous boundaries. The zero-shot results should therefore be interpreted mainly as evidence of a strong domain shift rather than as a direct ranking of the source models.

Fifth, BorDINO was evaluated only with $\lambda_b = 0.2$. No ablation was performed to study how the weight of the auxiliary boundary loss affects the two datasets. A smaller or larger value may change the balance

between region prediction and contour prediction, and the lack of improvement on Rabbani could partly depend on a suboptimal value of this hyperparameter rather than on the boundary task alone.

Finally, all models process frames independently, even though both datasets originate from surgical videos. The experiments do not use motion, temporal consistency, or information from adjacent frames. Such information could help distinguish transient reflections from actual blood, follow thin blood flows over time, and reduce unstable predictions between consecutive frames.

Future work should therefore tune the boundary-loss weight separately for each domain, evaluate boundary-guided feature fusion, and investigate temporal modelling or domain-adaptation strategies. In a boundary-guided extension, the predicted contour map could be used as a gate that reintegrates boundary features into the segmentation decoder, rather than serving only as an auxiliary training signal.

References

- [1] A. J. Miao, S. Lin, J. Lu, F. Richter, B. Ostrander, E. K. Funk, R. K. Orosco, and M. C. Yip, “HemoSet: The first blood segmentation dataset for automation of hemostasis management,” in *2024 International Symposium on Medical Robotics (ISMR)*, 2024.
- [2] N. Rabbani, C. Seve, N. Bourdel, and A. Bartoli, “Video-based computer-aided laparoscopic bleeding management: A space-time memory neural network with positional encoding and adversarial domain adaptation,” in *Proceedings of the 5th International Conference on Medical Imaging with Deep Learning*, vol. 172, pp. 961–974, 2022.
- [3] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015.
- [4] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, “UNet++: A nested U-Net architecture for medical image segmentation,” in *Deep Learning in Medical Image Analysis*, 2018.
- [5] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, “Encoder–decoder with atrous separable convolution for semantic image segmentation,” in *European Conference on Computer Vision (ECCV)*, 2018.
- [6] S. Yang, H. Wang, Z. Xing, S. Chen, and L. Zhu, “SegDINO: An efficient design for medical and natural image segmentation with DINO-V3,” *arXiv preprint arXiv:2509.00833*, 2025.
- [7] H. J. Lee, J. U. Kim, S. Lee, H. G. Kim, and Y. M. Ro, “Structure boundary preserving segmentation for medical image with ambiguous boundary,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 4816–4825, 2020.
- [8] T. Kataria and S. Y. Elhabian, “BoundarySeg: An embarrassingly simple method to boost medical image segmentation performance for low data regimes,” *arXiv preprint arXiv:2505.09829*, 2025.
- [9] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [10] L. Maier-Hein, A. Reinke, P. Godau, et al., “Metrics reloaded: Recommendations for image analysis validation,” *Nature Methods*, vol. 21, pp. 195–212, 2024.